Pre-analysis plan for a study of salience of study information and credibility assessment of key assumptions and inputs

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A note on this document. This pre-analysis plan was drafted with the assistance of Claude Code. It is unclear whether the survey will draw any responses at all. The plan was written for the optimistic case, in which enough people respond for a study to be conducted with the data. It was written now so that the analysis is fixed before any response is seen. **At this point, it is the opinion of the author that no human should spend much time reading this pre-analysis plan. It is only intended as a record.**

Status: the version registered on the AEA RCT Registry (the American Economic Association's registry of randomized controlled trials, trial 19997) on 2026-10-19, before the forms open. The structure follows the minimal elements of a clear hypothesis: population, interventions, arms, outcomes, hypotheses (contrasts, tests, alternative, estimation, standard errors), and heterogeneity stated separately and in full.

1. Question

Proposition 40 is a California ballot measure. It would tax Californians worth $1 billion or more once, at 5% of their net worth. Two academic studies estimate what the tax is worth to the state: Boll, Saez and Zucman (the BSZ analysis) and Jaros and Rauh (the JR analysis). One finds that the tax raises money for the state and the other that it loses money. This project is an Open Policy Analysis (OPA) of the two. Its website puts both estimates on one measure and shows, on a page called the comparison explorer, which inputs explain the gap between them. The project's survey asks respondents which of two proposed values of each disputed input (an input on which the two analyses use different values) they find more credible. The comparison explorer shows the same inputs, which study uses which value, and what each value does to the estimate. Most respondents will have seen that comparison on the site before they open the survey. This study asks whether the answers change when the comparison is restated at the moment respondents choose the input values, with each value labelled by its study.

2. Population and sample

- **Population.** Everyone who submits one of the four open forms (section 4.3) during the collection window (section 9), whether they reached the form through a project survey link or through a form address that someone forwarded (section 8). No one is screened out. Self-described experts and respondents without prior exposure are subgroups (section 6), not the

sample.

- **Unit of randomization.** The respondent, drawn at the click on a survey link (section 4.2).
- **Unit of analysis.** One submitted response.
- **Sample size.** The sample size is not controlled. It is whoever responds in the window. The number of responses depends on whether the launch of the OPA gains traction on social media, and it is likely that there will be none or very few. A reasonable range is 0 to 200. The staged rule of section 7 decides which analyses run once the realized number is known, and section 7 gives the detectable effects for a range of sizes.

3. Interventions

There are two manipulations. Both are delivered only in the treatment forms. Nothing is delivered to the control group, which makes it a pure control.

- **I1, the comparison page.** This is one page, titled "How the two estimates differ", shown after the opening questions and before the input questions. It opens with "Two academic studies" and gives each study on its own bullet with its month and its estimate (Boll, Saez and Zucman, the BSZ analysis, and Jaros and Rauh, the JR analysis). Neither bullet names a venue. Both are working papers of the National Bureau of Economic Research (NBER). Naming a venue for one study only would signal that it is the more reputable one, and that signal would become part of the treatment. It then gives the gap between the two estimates and names the common measure on which the explorer puts both. Next it splits the gap into the four terms of the comparison explorer's equation (a sum, one term per group of inputs, that adds up to the gap), largest first. Each term has a bar, its amount in billions and its share of the gap, and, for the first three, the values each study uses. A sentence names the two largest terms and their combined share. A two-sentence note says why the lost income tax is one term (explained after the table below) and that each amount depends on the order in which the inputs are changed. The full text is in Appendix A.
- **I2, the study labels.** In the input questions, each choice is labelled with its study ("BSZ:" or "JR:") instead of "A." or "B.". The choice text and order are otherwise unchanged.

The four terms, as generated at the launch commit (section 9), follow.[1] The gap is $145.8 billion.

Term on the page	Amount	Share of the gap
The income tax lost when billionaires leave: the share lost, how long it is counted, and billionaires' yearly income tax, together	-$81.3 billion	56%
Discounting the revenue for court risk	-$39.4 billion	27%
Billionaires who moved out in time	-$29.9 billion	21%
Other inputs together: the wealth on the list, avoidance, and three smaller inputs	+$4.9 billion	offsets 3%

[1] Each amount is the sum of the explorer's walk steps in that term (`comparison/export/r/dial_path.csv`). The walk moves the inputs one after another from the BSZ values to the JR values. The values come from `comparison/export/r/inputs.csv`. `comparison/survey/mock_forms.py` and `gas_build.py` generate the page from those files, so no number is typed. They also check that every walk step belongs to one term and that the terms add up to the gap.

The shares add to 101% because of rounding. A negative amount means that the term lowers the estimate as the inputs move from the BSZ values to the JR values, and a positive amount that it raises it. The page prints the first three amounts without a sign.

The page does not give one amount per survey question. The lost income tax is a product of the share lost, how long it is counted and the yearly amount. A split among those three factors would depend on the order in which they are changed, so it would misstate what any one of them does. The page shows them as one term and says why. As a result, the terms do not match the survey questions one to one. The first term covers three whole questions: the share lost, how long it is counted, and the yearly income tax. The question on billionaires who moved out in time spans two terms. The JR value (seven named moves) is the third term. The BSZ value (an allowance for unnamed movers, which is half of the BSZ analysis's allowance for avoidance) sits in the fourth term, together with the rest of that allowance. The two optional questions, on avoidance and on the wealth the tax applies to, also sit inside the fourth term. Only the court question matches one term. The small difference in discount rates is inside the first term and is not named, since the question on how long the lost income tax is counted (the horizon question, section 4.4) no longer asks about it.

4. Arms and assignment

4.1 Arms

Arm	I1	I2
Control (T = 0)	no	no
Treatment (T = 1)	yes	yes

I1 and I2 are delivered together. The design measures their joint effect and cannot separate them.

4.2 Assignment

Google Forms cannot randomize inside a form, so the draw happens at the click. Each survey link opens its entry point's control form or treatment form with probability 1/2, drawn in the browser (`Math.random() < 0.5`). The respondent sees nothing of either form before the draw. Draws are independent. Each entry point is a stratum: a group within which the draw is made, and which the analysis accounts for with its own indicator (section 6.1).

4.3 Entry points and forms

There are four forms: one control form and one treatment form for each of two entry points. The form a response lands in records its entry point and its arm. Respondents see neither label. Appendix B reproduces the control and treatment forms.

Entry point	Who arrives there	Control	Treatment
`explorer`	the survey button lower on the explorer page, and the overview page's two survey buttons	form 3	form 4

Entry point	Who arrives there	Control	Treatment
`share`	the share link (`comparison/site/survey/`): copied from the site, sent by email, or opened from the public posts through tinyurl.com/prop40-survey	form 5	form 6

Forms 1 and 2 belonged to a third entry point, `landing`: a panel shown to every visitor of the explorer before the results, inviting them to take the survey first. The panel was removed on 2026-10-17, before launch, because it stood between visitors and the explorer and could deter them from using it. Forms 1 and 2 were built but are never opened.

The OPA pages point readers to the comparison explorer several times, and the explorer shows the two competing estimates side by side. Every respondent who arrives through the `explorer` entry point has therefore seen the comparison before opening the survey. The `share` link opens a form directly, so its respondents are a mix: some have seen the estimates and some have not. For this reason the experiment mainly tests salience, that is, whether bringing that information back to mind at the moment respondents choose the input values changes the answers, and not the effect of a first exposure to it.

The `share` stratum also holds some visitors of the explorer and the overview. A survey button opened without a click event, through the context menu, a long press or a drag, goes to the share page, which draws and records in the `share` stratum (section 8). These visitors have seen the comparison. Their presence shrinks the difference that H2 looks for and does not touch H1.

4.4 Page order (all forms)

1. The title ("What are the actual quantities? Proposition 40, input by input") and the description, then two pre-treatment questions and the optional email:

    - "Do you consider yourself an expert in tax or fiscal policy?" (Yes / Somewhat / No; moved here from the end of the current survey, the version 1 forms that have been open since before the experiment)
    - "Before opening this survey, had you looked at news sources, web pages or academic studies discussing estimates of Proposition 40, the Billionaire Wealth Tax Proposal in California?" (Yes / No / Not sure; new)
    - "Your email (optional)" (moved here from the closing page on 2026-10-18, so it is asked before treatment; nothing is verified)

2. I1 (treatment forms only).

3. The page "The inputs" ("For each input, pick the value you find more credible."): the five required input questions, in three groups under plain section headings, in the order of the treatment page's first three terms (`GROUPS`, `regroup()` and `split_bundle()` in `mock_forms.py`):

    - "1. The income tax lost when billionaires leave": "The share of billionaires' income tax lost when people leave: which is more credible?", then how long the lost income tax is counted, then billionaires' yearly income tax. The three are ordered by each input's effect on the estimate, averaged over every order in which the explorer's dials (the controls that switch each input between its two values) can be turned, largest first.

- "2. Discounting the revenue for court risk": the court question.
- "3. Billionaires who moved out in time": "Billionaires who moved out of California before the 1 January 2026 residency date: which is more credible?" It spans two of the page's terms (section 3).

Each required question is followed by a free-text box, "Why? (optional)". These boxes carry the reasons that section 5.3 reports alongside S1.

Then comes the opt-in to the five smaller inputs, a required question with the choices "Yes, show me the five" and "No, skip them". Their page ("Five smaller inputs (optional)") follows it and asks about wealth that escapes the tax through avoidance and evasion, the wealth the tax applies to, non-citizen residents, the real-estate deduction, and the extra income tax from assets sold to pay the tax. One "Why? (optional)" box closes the optional page.

The current survey bundles departures, avoidance and the share lost into one question. In all the forms that question becomes three: the share lost and moved out in time (both required), and avoidance (optional). The two values offered in each of the three then concern one and the same input. The wealth question moves from the required questions to the optional page. I2 applies to all ten input questions in the treatment forms.

4. The closing page, "Anything else (optional)": missing inputs and support for Proposition 40. Every question on it is optional.

The question order and the three headings are the same in both arms, so they are not part of the treatment. They differ from the current survey's order. All the forms use one fixed choice order within each question, that of the current survey (version 1). In the first pretest (section 9) the second-listed value won the share-lost question (21 to 9) and the court question (18 to 4) in the control forms, so its language-model respondents showed no sign of favouring the first-listed value. That says little about people. Because the order is the same in both arms, an order effect cannot bias H1. It can shift the level of each question's answers in S3. The horizon question drops the discount rate (author, 2026-10-16): "How long the lost income tax is counted: which is more credible?", with the choices "For ever: billionaires who leave are assumed not to return." and "5 years, about the payment period, treating the departures as temporary." ("departures" replaced "responses" on 2026-10-17, because no respondent had been given a meaning for "responses"). The current survey's "Which field best describes your work?" is dropped (author, 2026-10-16).

Wording of the values (revised 2026-10-17, after the first pretest, and approved by the author the same day). In every input question the two values are the same kind of claim. Each gives a reason, or neither does. Neither value argues against the other. The texts are generated by `mock_forms.py` (`SHARE`, `DEPARTED`, `AVOID`, `INCOME`, `COURT`, `WEALTH`, `NONCIT`, `REAL`, `ASSET`), with every number read from the exports. What changed, and why:

- The share lost now states both values on the same base, "of all the income tax that billionaires pay" (`comparison/DISPUTES.md`, row 6): 45%, more than the 29% of wealth reported to have left, against 5%, the same as the 5% of wealth assumed to leave. Each comes with its analysis's own reason. The old pair ("45% of all billionaires' income tax, whoever leaves" against "5%, only the income tax of those who leave") was read in three ways in the first pretest: as a count of leavers, as shares of two different bases, and as losing the tax of those who stay. A respondent who accepts one analysis's count of the wealth that left but the other analysis's rule for which income tax is lost can now answer.

- The court question now asks whether to bring a court-risk discount into the estimate at all. Its title is "Discounting the revenue for the chance that courts strike the tax down". The two values are no discount, with the tax assessed at face value, against the wealth tax collected multiplied by the chance the tax survives, put at 50%. The discount applies to the wealth tax collected, not to the whole estimate (the JR analysis's chapter, PDF p.33). The comparison explorer likewise applies that chance (its symbol q) to the wealth tax only. The question drops "guesstimated", which all ten interviewees of the first pretest (section 9) flagged, and neither value sounds careless.
- The question on billionaires who moved out in time gives each value its own standard for counting a move. It no longer contains a rebuttal of the other value, or the third value (the "aggressive scenario") that interviewees read against both sides.
- The yearly income tax gives each value one assumption with its source, in the same words. The JR value names its two counts of top earners (212 and 500, read from the export's source note). The BSZ value names the study of top wealth holders. The old JR value carried a range and its two bounds, and the old BSZ value was one line ("halved"). In the first pretest 30 respondents chose the JR value and 3 the BSZ value, and six of ten interviewees named the difference in length as a reason.
- Four of the five optional questions give both values the same form ("Taxed" and "Not taxed", "Counted" and "Not counted"). In the first pretest the bare JR values ("None.", "Not counted.") drew 0 or 1 choices each. The avoidance question keeps its first-pretest text ("5%: a small allowance argued from enforcement." against "None."), the author's choice. The BSZ value of the wealth total now says how much of that total belongs to the non-US citizens asked about in the next question.
- The description no longer promises "the reason given for each" value, says "most of these are empirical quantities", and says 5 to 10 minutes (careful readers took a median of 14 in the first pretest). Its first paragraph ends with a sentence of motivation: the answers help get closer to agreeing on common facts for policy discussions. The opt-in says three things. First, the next five questions explain up to a stated share of the difference between the analyses. That share is the net of the three walk steps that hold the five inputs (the list, avoidance, and the three minor inputs as one step), divided by the gap, and it is the explorer's "other" term. Second, answering them makes the survey a stated share longer. That share is the number of words in the optional section divided by the number of words in the main part. Third, answering them is optional. The email question is "Your email (optional)": nothing is verified. After submitting, every form shows the same confirmation: thanks, the motivation sentence, and two short links, tinyurl.com/prop40-explorer and tinyurl.com/prop40-opa.

5. Outcomes

5.1 Coding of one answer

For respondent i and required question k ($k = 1$ to 5, in the order the forms ask them: the share of billionaires' income tax lost; how long the lost income tax is counted; billionaires' yearly income tax; discounting the revenue for court risk; billionaires who moved out in time):

- $y_{ik} = +1$ if the answer is the BSZ value,
- $y_{ik} = -1$ if the answer is the JR value,
- $y_{ik} = 0$ if the answer is "About equally credible" or "Unsure".

The two neutral answers mean different things. "Unsure" is a don't-know: the respondent cannot judge. "About equally credible" is a judged tie: the respondent has judged and neither value wins. Both are offered because dropping an explicit don't-know would push respondents without a view onto the middle answer. Both score 0 in the lean index (section 5.2), and the secondary outcomes S1 and S2 (section 5.3) report them separately.

In the treatment forms the label gives the study. In the control forms the choices are labelled A and B, in the choice order of the current survey (version 1). A is the JR value and B the BSZ value in the share-lost, horizon, yearly-income-tax and moved-out questions. In the court question A is the BSZ value (no discount) and B the JR value. In the five optional questions A is the BSZ value. A private tab named Key in the spreadsheet that collects the responses carries the same mapping, and Appendix B shows both forms. All five questions are required, so no value is missing. The avoidance and wealth questions are now optional (section 4.4), so they are not among the five. In the share-lost question both values are shares of the same total, the income tax billionaires pay, each shown with its analysis's reason.

5.2 Primary outcome

The lean index. For respondent i, $\text{Lean}_i = (y_{i1} + y_{i2} + y_{i3} + y_{i4} + y_{i5})/5$, the mean of the five coded answers of section 5.1. It runs from -1 (the JR value on all five) to $+1$ (the BSZ value on all five). The primary outcome is the index, not any one of its inputs.

5.3 Secondary outcomes

- S1. The share of the five questions answered "Unsure".
- S2. The share answered "About equally credible".
- S3. Each of the five y_{ik}, coded as in section 5.1, as the outcome of the H1 equation. For each question, the shares of the three answers (the BSZ value, the JR value, a neutral answer) are also reported by arm, descriptively.

In the first pretest, "About equally credible" was 2% of the required answers and "Unsure" 45%. The reasons given with "Unsure" on the horizon and court questions often said that neither value was right ("100% is too high, but 50% is a guess"). S1 therefore mixes not knowing with rejecting both values. The author keeps the two answers as they are (2026-10-17). The reasons given with "Unsure" are reported alongside S1.

The five optional smaller-input questions, avoidance and the wealth base among them, are excluded from every registered outcome, because only respondents who opt in answer them. Their answers are reported descriptively, with the share of each arm that opted in.

6. Hypotheses and estimation

6.1 Main hypothesis (primary)

H1. $E[\text{Lean} \mid T = 1] - E[\text{Lean} \mid T = 0] = 0$, against the two-sided alternative that it is not 0. In words, the mean lean index in the treatment arm equals the mean in the control arm.

H1 is estimated by ordinary least squares (OLS), a linear regression of the lean index on the treatment indicator and an indicator for the entry point (the stratum of section 4.2):

$$\text{Lean}_i = \alpha + \beta\, T_i + \gamma\, E_{\text{share},i} + \varepsilon_i$$

Here T_i is 1 if respondent i was drawn into the treatment arm and 0 if into control. $E_{\text{share},i}$ is 1 if the response landed in a `share` form and 0 if in an `explorer` form (section 4.3). α, β and γ are coefficients to be estimated, and ε_i is the error term. H1 is $\beta = 0$. Standard errors are heteroskedasticity-robust (HC2). Such standard errors allow the lean index to vary more in one arm or entry point than in another, and the HC2 variant corrects for small samples. The test is at the 5% level. The p-value uses the t distribution with degrees of freedom equal to the realized number of responses, N, minus the number of regressors. Below 30 responses the p-value of H1 comes from randomization inference instead (section 7). The estimate is an intention-to-treat effect, because whether a treated respondent read the comparison page is not observed. H1 is reported with the control-arm mean, the coefficient, its HC2 standard error, the 95% confidence interval, the p-value, N by arm and entry point, and the minimum detectable effect at the realized N. A second estimate, reported for robustness, adds to the H1 equation indicator variables for the two pre-treatment answers (for expertise, "Yes" and "Somewhat", with "No" as the base; for prior exposure, "Yes" and "Not sure", with "No" as the base) and the after-the-vote indicator of section 9, which is dropped if every response was submitted before the vote. The robustness estimate is reported alongside the main estimate and does not replace it.

Secondary outcomes S1 to S3 are estimated with the same equation. For S3 the outcome is y_{ik}. The five S3 tests form one family, adjusted with the Romano-Wolf step-down procedure (section 6.3). S1 and S2 are one test each. They are not adjusted, and they are run at any N together with H1. In their own standard-deviation units their minimum detectable effect is H1's.

6.2 Heterogeneity (all secondary)

Every heterogeneity test below is secondary. None changes the conclusion on H1.

The plan declares D = 3 candidate dimensions, that is, respondent characteristics along which the effect may differ. All are binary and recorded before treatment: entry point (H2), prior exposure (H3) and expertise (H4). Whether a respondent gives an email is asked before treatment, on page 1, but it is not registered as a dimension, so it stays descriptive and is not one of the three. No other dimension is used unless it is registered before the data are seen, and adding one enlarges every family below.

H2 to H4 are the one-dimension analyses. With H5 below, they form one family of four tests ($\binom{3}{1} = 3$ one-dimension tests, one per dimension, and H5), adjusted together with the Romano-Wolf procedure. Each of H2 to H4 tests one interaction coefficient, estimated in the H1 equation with the subgroup indicator G_i and its interaction with T_i added:

$$\text{Lean}_i = \alpha + \beta\, T_i + \delta\, (T_i \times G_i) + \lambda\, G_i + \gamma\, E_{\text{share},i} + \varepsilon_i$$

G_i is the subgroup indicator defined under each hypothesis below, λ is its coefficient, and δ is the difference between the treatment effect inside the subgroup ($G_i = 1$) and outside it ($G_i = 0$). The test is $\delta = 0$ against $\delta \neq 0$, with HC2 standard errors.

- **H2, entry point.** $G_i = 1$ if the response landed in an `explorer` form, 0 if in a `share` form. Prediction: the effect is smaller in absolute value for `explorer`, whose respondents have all seen the comparison on the page. Some `share` respondents have not. The entry point is not

randomized. Differences in levels between entry points therefore describe who arrives where. Within each entry point, T is randomized. For H2, G_i is the entry-point indicator itself, so λG_i is dropped, because it would duplicate $E_{\text{share},i}$. Since β is the effect for `share`, the prediction is that δ has the opposite sign to β and is smaller than $2|\beta|$ in absolute value. The test stays two-sided.

- **H3, prior exposure.** $G_i = 1$ if the answer to the prior-exposure question is "No", 0 if "Yes" or "Not sure". Prediction: the effect is larger in absolute value for G = 1, so δ has the same sign as β. The test stays two-sided. The answer is self-reported, and the question does not name the project's own pages. An `explorer` respondent who answers "No" has seen the comparison on the page and still counts as G = 1, by this rule. The cross-tabulation of the answer by entry point is reported with H3, so the reader sees how many such respondents there are.
- **H4, expertise.** $G_i = 1$ if the answer to the expertise question is "Yes" or "Somewhat", 0 if "No". No direction is predicted.
- **H5, experts without prior exposure.** The subgroup is the respondents who answer "Yes" or "Somewhat" to the expertise question and "No" to the prior-exposure question. H5 is that the treatment effect on the lean index inside this subgroup is 0, against the two-sided alternative. It is estimated with the H1 equation on the subgroup's responses only, with HC2 standard errors. These respondents know the field but had not seen estimates of Proposition 40, so for them the treatment page is closest to a first exposure. H5 is tested together with H2 to H4, in one Romano-Wolf family of four tests, and is run only once the number of responses N reaches that family's threshold (section 7).

Analyses that cross two or three dimensions are staged secondary analyses (section 7). Crossing k dimensions gives 2^k cells. The k-way difference of effects is the coefficient on $T \times G_1 \times \cdots \times G_k$ in the saturated version of the H1 equation (the equation with the k subgroup indicators, all their products, and the product of each with T added), with HC2 standard errors. The order of an analysis is the number of dimensions it crosses. Each order is its own Romano-Wolf family over every combination of the three dimensions: $\binom{3}{2} = 3$ two-way differences, and one three-way difference. The effects inside the cells form their own families: 6 for one dimension (3 dimensions x 2 halves), 12 for two (3 pairs x 4 cells), and 8 for three. The effect inside a cell is the coefficient on T in the H1 equation estimated on that cell's responses only, with HC2 standard errors. When the cell is defined by the entry point, the entry-point indicator is constant inside it and is dropped. A further staged family applies the same estimator to each of the five y_{ik} inside each half of each dimension: 5 questions x 3 dimensions x 2 halves, 30 tests in one Romano-Wolf family (tier 5 of section 7).

Both subgroup questions are asked before the treatment page, so treatment cannot change them. Support for Proposition 40 stays on the closing page, after treatment, and is not used for any registered test (decided 2026-10-17, because asking it first would put the vote in respondents' minds, in both arms, before they judged the values). It is optional, so the share who answer it is reported by arm alongside its descriptive results.

6.3 Multiple testing

Every family of secondary tests is adjusted with the Romano-Wolf step-down procedure (Romano and Wolf 2005, computed with the algorithm of Clarke, Romano and Wolf 2020). The families are: the five questions of S3; the one-dimension interactions H2 to H4 together with H5; the three

two-way differences; the effects inside cells, one family per order; and each of the five questions inside each half of each dimension (30 tests, tier 5 of section 7). The three-way difference is a single test and is not adjusted. The procedure keeps the chance of at least one false rejection in the family at 5% (the family-wise error rate). It does so by comparing each test statistic with the bootstrap distribution of the largest statistic in the family. Tests that move together therefore cost less than under the Holm or Bonferroni corrections, which treat every test as if it were independent. The five questions and the index are correlated, so the saving matters here. The statistics are the HC2 t-statistics of the registered equations. The bootstrap resamples respondents with replacement within each entry-point stratum. It uses 2,000 draws. The seed is 20261017, the seed of the second pretest's analysis (section 9), set once in the analysis script before the first family is drawn, so that the draws are reproducible and not chosen after the results are known. The power appendix states the thresholds under this procedure and the Bonferroni bound they sit below.

7. Power and staged analyses

The number of responses is unknown. The expected range is 0 to 200 (section 2). The plan therefore has one main hypothesis with one outcome (H1, the lean index), tested whatever the number of responses. Further analyses are run only once the realized number of responses N reaches the threshold at which each can detect a standardized effect of 0.5 SD with 80% power. A standardized effect is the effect divided by the standard deviation (SD) of the outcome in the control group. 80% power means an 80% chance of rejecting the null hypothesis when the true effect is of that size. The target of 0.5 SD is a medium effect by the usual convention. The thresholds, their formulas and simulations of the registered estimator are in the power appendix, `comparison/survey/power/power.pdf` (built from `power.tex` and `sim.py`, which generates every number in it).

H1 is estimated whenever at least one entry-point stratum has responses in both arms. A stratum with responses in one arm only is dropped from the H1 equation, and the count dropped is reported. If only one stratum remains, the entry-point indicator is dropped with it. Below 30 responses, the smallest N at which the appendix verified the test's level, the p-value of H1 comes from randomization inference in place of the HC2 t-test. The treatment indicator is re-assigned at random within each stratum 10,000 times, keeping each stratum's realized number of treated responses, with the seed of section 6.3. The p-value is the share of re-assignments whose HC2 t-statistic for β is at least as large in absolute value as the realized one. The estimate and its HC2 confidence interval are reported as well.

The table below summarizes the appendix's Table 1. It gives the responses needed for 80% power against an effect of 0.3 SD, with the Romano-Wolf step-down applied within each family. It assumes that 60% of responses come through `explorer` and 40% through `share`, and that expertise and prior exposure each split the responses in half. "RW" is the simulated Romano-Wolf threshold. "Bound" is the closed-form Bonferroni bound, which the RW threshold sits below. A dagger marks a family whose tests are nearly independent, where the simulation found no threshold below the bound and the bound is used. For the five questions, column A is 0.3 SD on a question and column B is the smaller question-level effect (about 0.18 SD) that goes with 0.3 SD on the index. For heterogeneity, column A is the effect inside each cell and column B is the highest-order difference of effects. Each family covers every combination of the three dimensions.

	Analysis	A: tests	A: RW	A: bound	B: tests	B: RW	B: bound
1	Main hypothesis H1, the lean index	1	349				
2	H1 by outcome, each of the five questions	5	440	520	5	1,110	1,410
3	Heterogeneity, one dimension at a time (3 combinations, 2 cells each)	6	920	1,346	4	1,983 (dagger)	1,983
4	Heterogeneity, two dimensions at a time (3 combinations, 4 cells each)	12	2,010	3,054	3	7,090	7,445
5	Heterogeneity, three dimensions at a time (1 combination, 8 cells)	8	3,850	5,684	1	23,256	

The staged rule follows (the appendix's Table 4). It is set at the registered target of 0.5 SD, with the same shares. The Romano-Wolf threshold governs. The bound is given for reference.

Tier	Analysis	Tests	Responses needed (RW)	Bound
0	H1, the lean index, with S1 and S2	1	always run; detects 0.5 SD from 126	
1	Each of the five questions	5	150 (about 400 for an effect matching 0.5 SD on the index)	187 (497)
2a	One dimension at a time: the effect inside each half (expertise, prior exposure, entry point; email descriptively)	6	330 (540 if one half holds a quarter)	485 (776)
2b	The difference between the halves (H2 to H4) and H5, experts without prior exposure	4	714 (dagger) (950 if one half holds a quarter)	714 (952)
3a	Two dimensions at a time: the effect inside each of four cells	12	740	1,100
3b	Two dimensions at a time: the two-way difference	3	2,681 (dagger)	2,681
4a	Three dimensions: the effect inside each of eight cells	8	1,380	2,047
4b	Three dimensions: the three-way difference (one test)	1	8,373	
5	Each question inside each half, one dimension at a time	30	390 (about 1,070 for an effect matching 0.5 SD on the index)	509 (1,351)

The 2b threshold assumes the H5 subgroup holds a quarter of the responses. If it holds half, H5 alone would need 357, and H2 to H4 set the threshold. Explorer respondents have all seen the

comparison, so the subgroup will sit mostly in the `share` stratum. The rule is applied at the realized shares.

Tier 1 runs from 150 responses. Between 150 and about 400, the number a question-level effect that matches 0.5 SD on the index needs, the five questions are reported as estimates with 95% confidence intervals and adjusted p-values, not read as tests of a common effect. Tier 5 runs from 390 and is read the same way below about 1,070.

Heterogeneity is unlocked one order at a time. Two-dimension analyses are run only if the one-dimension analyses of the same kind are run, and three-dimension analyses only if the two-dimension ones are. Each added dimension multiplies the responses a difference of effects needs by about four, because the cells halve in size and the difference adds cells.

The rule uses only N and the realized shares of pre-treatment subgroups at the close, so it does not depend on the results. At the close, before any outcome is tabulated, `sim.py` is rerun once with the realized shares in place of the assumed ones (`P_EXPL` and `SHARES0`, which hold 60/40 for the entry point and 50/50 for expertise and prior exposure, and `P_H5`, which holds the H5 subgroup's quarter), and the regenerated `out/thresholds.tex` is committed together with the realized shares. Those thresholds decide which analyses run. An analysis below its threshold is reported as "not run: below its threshold", with the threshold. Until then the registered appendix (`out/summary.tex` and `out/thresholds.tex`) governs. No other regeneration is foreseen. Any other change to the appendix after registration is a deviation and is logged (section 9). The appendix's simulations use 199 bootstrap draws inside each simulated sample, fewer than the registered 2,000, so its thresholds err on the high side.

In the lean index's own units, with the simulated index SD of about 0.5 (an assumption, not a measurement), 0.5 SD is a change of 0.25 in the mean index. That is, for example, 125 one-step answer moves per 100 treated respondents, or a quarter of treated respondents moving all five answers one step. A step takes one answer from neutral to a study's value, or from the other study's value to neutral. 0.4 SD is 0.20 (100 moves per 100, or 20% moving all five answers). 0.3 SD is 0.15 (75 moves per 100, or 15%). 0.2 SD is 0.10 (50 moves per 100, or 10%). 0.1 SD is 0.05 (25 moves per 100, or 5%). The appendix's Table 3 and Figure 1 give these, with the responses H1 needs at each size. At the expected numbers of responses the main test detects 0.79 SD with 50 responses, 0.56 with 100 and 0.40 with 200.

8. Checks, attrition and exclusions

- **Sample.** The sample is every response submitted to the four open forms (3 to 6) in the window. Two kinds of response are excluded: test responses made before launch (deleted from the sheet), and responses to the two current forms, which predate the experiment and are reported on their own.

- **Attrition.** Google Forms records only submitted responses. If the comparison page makes people quit, the treated respondents would differ from the controls. Three checks are reported whatever the result:

    1. Within each entry point, the number of submissions per arm is compared with an even split (exact binomial test).
    2. The two pre-treatment answers are compared across arms (chi-squared tests).
    3. A count of draws is kept. Each click on a survey link sends one request, at the draw,

carrying only the entry point and the arm (no identifier, no cookie). The explorer's notes disclose the count in one sentence ("This page records, without any identifying data, that a survey link was opened and which form it drew."). The sentence is shown before the draw and to both arms. The registered attrition check is submissions divided by draws, by arm and entry point. The binomial test above is the fallback if the count fails. With 100 submissions the binomial test detects only a split of about 60 to 40. A respondent who quits on the comparison page leaves no other trace.

Draws count clicks, not people. A respondent who clicks twice is two draws, a program that loads the share page and runs its script is a draw with no respondent, and a browser that blocks the request is a submission with no draw. None of these depends on the arm, so the comparison of the ratio across arms stands. The ratio itself is not a response rate. The count fails for an entry point, and the binomial test is then its only attrition check, if the count recorded no draw on a day on which that entry point's forms received submissions, or if either of its forms has more submissions than draws (the sign of a forwarded address, below).

If the checks show differential attrition, H1 is reported as registered and also with the trimming bounds of Lee (2009). Differential attrition means that, within an entry point, the two arms' submissions per draw differ at the 5% level in a two-sided test of two proportions, or, when the count fails, that the exact binomial test of item 1 rejects at the 5% level. Within each entry point, the arm with the higher submission rate q_H is trimmed by the share $(q_H - q_L)/q_H$ of its responses, where q_L is the other arm's rate. It is trimmed once from the top and once from the bottom of its lean-index distribution. The two trimmed estimates of the H1 equation bound the effect among respondents who would have submitted in either arm. When the count fails, the submission counts stand in for the rates, since the draw is even in expectation. The bounds are reported without a test when an arm has fewer than 50 responses in the entry point.

- **Duplicates.** Of responses with the same email (compared after trimming spaces and lowercasing), only the earliest by timestamp is kept. Anonymous duplicates cannot be detected. One person can land in both arms, which shrinks any measured effect, because that person's answers then appear on both sides of the comparison. The forms do not require a Google sign-in, the only way Google Forms limits one account to one response, because a sign-in would deter respondents and record their accounts. The confirmation screen shows no "Submit another response" link.

- **Bulk and automated submissions.** Google Forms does not screen out automated submissions. A bot that posts to one form's address skips the draw at the click, so it fills one arm only. That would bias the estimate, not only add noise. The rule is fixed before launch. Three or more submissions to the same form whose timestamps all fall within one span of ten minutes, with identical answers to every choice question, count as one submission, and only the first is kept. The number removed per form is reported. H1 is also reported with no response removed. A burst in one arm also shows in the per-arm count check above. A bot that varies its answers is not caught by this rule. As a robustness check, H1 is also reported without the submissions of every clock hour (Pacific time) in which one form received five or more submissions while its pair's form received none. The hours and forms removed are reported.

- **Respondents who give an email.** Their answers are also reported separately, as the survey promises. The registered analysis uses all responses.

- **Links that skip the draw.** The share page sends the browser to the form's own address, which a respondent can copy and forward. Everyone who opens a forwarded form address lands in one arm. Without JavaScript, every survey button and the share page itself lead to one form, the control form of the `share` pair (form 5). The share page's fallback link names that form, and the treatment form's address sits only inside the page's script. A program that reads the page without running the script, such as a crawler or a form-posting bot, finds the same one address. These responses are not a random half. They add control responses to the `share` stratum, with no draw. Neither group can be identified in the data. The Copy link button copies the share page's address, not a form's address. The public posts no longer ask readers to pass on the project link rather than the form's address (section 11). A survey button opened without a click event also goes to the share page and draws there (section 4.3; `comparison/R/site.R`, `survey_link()` and `survey_draw_js()`). Those respondents count as `share` respondents. The entry point in the data is the form a response lands in, not the button pressed. Submissions above draws for one form are the sign of a forwarded address, of the no-script path or of a bot. For form 5 all three point the same way.

9. Timeline, registration and priors

- **Registration.** The plan is registered on the AEA RCT Registry, trial 19997, before the four forms open, with `comparison/survey/plan/plan.pdf` and `comparison/survey/power/power.pdf` attached.
- **Ethics.** No IRB review was sought, and the registry's IRB field says so. Answers are anonymous unless the respondent gives an email, which is optional and is used as section 8 says, to remove duplicates and to report that group separately. The forms do not say that two versions exist. Neither version says anything false, since the treatment page restates numbers from the project's own pages.
- **Launch.** The forms open together with the site change. The launch commit is recorded here on launch day (step 6 below): ea8995df886a3a1994bdebd8c8397746c69102e5. The registry's hidden design details and its repository note carry the same hash.
- **Close.** Collection closes on 4 November 2026 at 23:59 Pacific time, when the four forms stop accepting responses. Responses with a later timestamp are not part of this analysis. Responses submitted after the vote of 3 November 2026 and before the close are included. The time of submission is recorded, and an indicator for "after the vote" (a submission timestamp after 20:00 Pacific time on 3 November 2026, when California's polls close) is a covariate in the robustness specification of H1 (section 6.1), reported alongside the main estimate.
- **Pretest.** On 2026-10-17, 100 simulated respondents (language-model agents given personas, entry paths and arms) took mock versions of the six forms, and ten of them were interviewed (`feedback/008-2026-10-17-rct-pilot/`). This plan calls it the first pretest. It tested wording and comprehension only. Its answers are not data for this study. The question texts were revised after it (section 4.4). The second pretest, the same day, with 100 new simulated respondents and no interviews, ran the revised forms through this plan's analysis as it stood on 2026-10-17, which still included the later-dropped secondary outcome S4 (`feedback/009-2026-10-17-rct-pilot-2/`). Its answers are not data either.
- **Analysis code.** `comparison/survey/pilot_analysis.py` runs this plan's analysis. Its version at the launch commit is the registered one. A later change is a deviation and is logged.

- **No interim analysis.** Before the close, only response counts per form are looked at.
- **Report.** The analysis is written up by 31 December 2026, the trial end date given to the registry.
- **Data.** After the close, the responses, with the email column and the free-text answers removed, and the draw count are published in the repository with the analysis. The free-text answers are never published. Where section 5.3 reports them, the analysis summarizes them without quoting them.
- **Priors.** No forecasts of the effect are collected.
- **Deviations.** Any change after registration is logged at the end of this file with its date and reason.

How to launch.

1. The six forms exist: `buildExperiment()` in the Google Apps Script program that `comparison/survey/gas_build.py` writes built them on 2026-10-17, named "exp: Prop 40 survey (ENTRY, ARM)", with the entry point and the arm in place of ENTRY and ARM, and its `fixEnding()`, `moveEmail()` and `fixPrior()` edited them in place since (forms 1 and 2, for the dropped `landing` entry point, are never opened). On launch day, run `fixPrior()` once more: it leaves a finished form unchanged and logs, for every copy of each form, its edit and respondent links, whether it is finished and its item count, in the format `fill_links.py` reads. A WARNING line in that log means two copies share a name: trash the unfinished one (finished: false) by hand before step 2. `buildExperiment()` is run again only if a form is missing. Each form takes about 110 seconds and a run stops at six minutes, so a run builds three forms; run it again. It trashes a half-built copy and rebuilds it, skips a finished form, and logs the links only of the forms it builds in that run.
2. Fill `comparison/data/survey-links.csv` from that execution log: `python3 comparison/survey/fill_links.py LOG.txt` writes the four rows (`entry`, `arm`, `url`), each form's respondent link in the row of its entry point and arm, in place of the placeholder file and its comment line. It refuses a log that lacks one of the four forms, names two copies of one, or shows a form as unfinished. Pasting by hand works too: each respondent link in its row, and the comment line about placeholders deleted.
3. The thank-you screen after Submit carries the two short links (tinyurl.com/prop40-explorer and tinyurl.com/prop40-opa). `fixEnding()` adds them to forms built before the afternoon of 2026-10-17. No link sits before Submit, because a control respondent could open it, go Back and change answers.
4. Deploy `comparison/survey/draws.gs` as a web app (execute as me, anyone can access), as its header says. Paste its URL into `comparison/data/survey-config.csv` as `draws_url`.
5. Rebuild with `Rscript comparison/run.R`, then run `Rscript comparison/tests/testthat.R`.
6. Commit. Record that commit's hash under "Launch" above, in a second commit (a hash cannot sit inside its own commit). The registry takes the same hash in step 7.
7. Register on the AEA RCT Registry, with the current `comparison/survey/plan/plan.pdf` and `comparison/survey/power/power.pdf` attached in place of any older copies, and the launch commit's hash in the "Experimental Design (Hidden)" field and in the repository note under "Docs & Materials". Then link each of forms 3 to 6 to a spreadsheet (the form's Responses tab, "Link to Sheets"), because the builder makes each form as a Drive copy and a copy keeps no spreadsheet link. The control forms' A and B are mapped to the two studies in section 5.1. Then publish forms 3 to 6 (each form's "Publish" button, with responses

accepted), because the forms are built unpublished and closed, and push. The push puts the four forms' links on the site. Once GitHub Pages has deployed it (a few minutes), close the two current forms, the version 1 and version 2 forms whose links are the placeholders in `survey-links.csv` (section 11: they close when the new forms open).

10. Limits

- **Who answers.** The results describe people who take the survey, not experts in general. The randomization makes the two arms comparable, not representative. Respondents are recruited through the project's own pages, which say the survey is primarily for economists and policy analysts, and through the author's posts on LinkedIn and Bluesky and the emails that pass the share link on. The sample is the author's network and whoever those posts reach.
- **Prior exposure.** Every `explorer` respondent, and many `share` respondents, will have seen the comparison before the survey. The treatment therefore restates information most respondents already have, which likely shrinks the effect. H2 and H3 are there to show how much. Exposure can also happen during the survey, or between respondents. A control respondent can open the explorer in another tab while answering, and a treated respondent can describe the comparison page to a colleague who then takes the survey. Neither is visible in the data. Both move the control arm toward the treatment and shrink the effect.
- **A bundled treatment.** I1 adds how much of the gap each of four groups of inputs explains. I2 adds which study holds each value. Naming the studies may move answers through views of the authors or of the tax, not only through the numbers. H1 measures the two together.
- **Terms and questions differ.** The page's four terms do not match the five questions one to one (section 3). Only for the court question does the page give that input's own amount. A question-level effect (S3) therefore cannot be read as a response to that question's amount on the page.
- **Neutrality.** The control forms keep the survey's rule of naming neither analysis. The treatment forms name both, by design.
- **The sides can be told apart without labels.** In the first pretest, each persona's view of wealth taxes predicted its lean index as strongly in the control forms as in the treatment forms. On average, the lean index of the personas who favour wealth taxes and that of the personas who oppose them differed by 0.68 in the control forms and by 0.71 in the treatment forms. The reason is that a value's content shows which side it helps. Respondents from the explorer have also seen each value's study on the explorer's dials. The explorer's tips also grade the evidence behind each value (data, research, guesswork, scenario or derived), a judgment of this OPA's author. Respondents from the explorer in both arms have seen those grades, so the `explorer` stratum's control answers are not those of respondents who met the values without labels or grades. The labels may therefore add less than their presence suggests, which shrinks the effect that H1 can find.
- **What the entry texts promised.** The drafts of the public post that carries the share link said the survey "names neither analysis", which is false for the treatment forms. In the first pretest, ten of the eighteen simulated respondents who came by that link to a treatment form noticed the contradiction. The drafts were edited on 2026-10-17, before posting: no promise about naming, "5 to 10 minutes", and a request to pass on the project link rather than the form's address. That request was removed on 2026-10-18 (section 11).

11. Decisions taken

The decisions that were open in earlier drafts were settled on 2026-10-17. Each is listed with the section that records it.

- The two current forms close when the new forms open.
- The overview page's survey buttons send to the `explorer` pair of forms.
- The value texts were revised after the first pretest (section 4.4).
- The treatment page was fixed (Appendix A).
- The bulk-submission rule, the draw count and the links that skip the draw were set (section 8).
- The close date and the treatment of post-vote responses were set (section 9).
- The Romano-Wolf adjustment was adopted (section 6.3).
- The two neutral answers are kept as they are (section 5.1).
- The public post was edited before posting (section 10).
- The support question is kept after treatment (section 6.2).

Later on 2026-10-17, before launch, the author made four more decisions.

- The `landing` entry point and its panel are dropped, so forms 1 and 2 are never opened and the experiment is framed as a test of salience (sections 1 and 4.3).
- The expected number of responses is 0 to 200 (section 2).
- The expert respondents without prior exposure form a registered subgroup (section 6.2).
- The share link keeps opening a form, so that some respondents may not have seen the estimates.

On 2026-10-18, before launch, the author moved the email question from the closing page to page 1, after the two pre-treatment questions (section 4.4). Whether a respondent gives an email stays a descriptive variable, not a registered dimension (sections 6.2 and 8). The form's description no longer says that the email comes near the end, and the opt-in's stated extra length fell from 60% to 55%, because the main part now includes the email question. The same day, the author dropped the former secondary outcome S4 (the mean absolute lean index). The secondary outcomes are S1 to S3 (section 5.3).

Also on 2026-10-18, before launch, the author removed from the public posts (LinkedIn and Bluesky) the request to pass on the project link rather than the form's address (sections 8 and 10). A respondent who forwards a form's address sends the people who open it to one arm, with no draw. The sign of this in the data is unchanged: a form with many submissions and few draws (section 8).

On 2026-10-19, before launch, the author settled the open items of the plan's review (`feedback/011-2026-10-18-pap-review/`).

- S3 is one family of five tests on the coded answers (sections 5.3 and 6.1).
- Differential attrition is met with the bounds of Lee (2009) (section 8).
- A stratum with responses in one arm only is dropped, and below 30 responses the p-value of H1 comes from randomization inference (section 7).
- A robustness check removes hours in which only one form of a pair received a burst of submissions (section 8).

- The responses are published after the close without the email column and the free-text answers, which are never published (section 9).
- The site's fallback for browsers without JavaScript stays as it is (section 8).
- The title stays as it is.

Appendix A. Treatment page text

The page is generated by `python3 comparison/survey/gas_build.py ITEMS.json SOURCE_FORM_ID` (the Apps Script that builds the forms) and previewed by `python3 comparison/survey/mock_forms.py ITEMS.json OUT_DIR`. The text comes from `STUDIES_LEAD`, `STUDIES`, `INTRO`, `TERMS`, `NOTE` and `CLOSE` in `mock_forms.py`. The page has a title, a lead sentence ("Two academic studies estimate what Proposition 40 is worth to California:") with one bullet per study, its month and its estimate, two paragraphs (one on the gap and the common measure, one on the two largest terms and their share), the four terms of section 3 with the BSZ and JR values of the first three, a two-sentence note (why the lost income tax is one term, and that each amount depends on the order in which the inputs are changed), and one closing sentence. The full text, as generated at the launch commit, follows.

How the two estimates differ

Two academic studies estimate what Proposition 40 is worth to California:

- Boll, Saez and Zucman (BSZ, August 2026): +$106.8 billion.
- Jaros and Rauh (JR, September 2026): −$39.0 billion.

The gap is $145.8 billion. Both are on one measure: the wealth tax collected, plus the extra income tax from assets sold to pay it, minus the income tax lost when people leave.

Two parts explain 83% of the gap: the income tax lost when billionaires leave, and the chance the tax survives in court. Below, how much of the gap each part explains, with each study's values for the first three.

1. **The income tax lost when billionaires leave: the share lost, how long it is counted, and billionaires' yearly income tax, together**: $81.3 billion, 56% of the gap. BSZ: 5% of all the income tax that billionaires pay is lost; counted for 5 years; $3.06 billion a year. JR: 45% of all the income tax that billionaires pay is lost; counted for ever; $4.55 billion a year.
2. **Discounting the revenue for court risk**: $39.4 billion, 27% of the gap. BSZ: no discount. JR: a 50% discount.
3. **Billionaires who moved out in time**: $29.9 billion, 21% of the gap. BSZ: no billionaire counted by name (its allowance for 5% of the wealth moving away in time is in part 4). JR: seven billionaires counted on public reports of their moves.
4. **Other inputs together: the wealth on the list, avoidance, and three smaller inputs**: +$4.9 billion, offsetting 3% of the gap.

The income tax lost is shown as one amount because its three inputs multiply each other. For the same reason, each amount depends on the order in which the inputs are changed.

The questions that follow ask which study's value of each input you find more credible.

Appendix B. The two forms

The pages that follow reproduce the control form (form 3) and the treatment form (form 4) as respondents meet them, page by page, with the confirmation message shown after Submit. Each copy is printed from the form's own definition as Google Forms served it in preview on 18 October 2026 (`comparison/survey/plan/forms/control.json` and `treatment.json`, printed by `print_forms.py` in the same folder). Forms 5 and 6, the `share` entry point's forms, carry the same text as forms 3 and 4. Only Google's internal item ids differ.

Appendix B.1. The control form (forms 3 and 5)

Printed from the form's own definition as Google Forms served it in preview on 18 October 2026. Each page of the form starts at a purple rule. Respondents see the text exactly as below, one page at a time.

PAGE 1 OF 4

What are the actual quantities? Proposition 40, input by input

Proposition 40 would tax Californians worth $1 billion or more once, at 5% of their net worth. Estimates of what it is worth to the state depend on a handful of inputs. Each question shows one input and two values that have been proposed for it. Tell us which value you find more credible. Your answers will help us get closer to agreeing on common facts to inform important policy discussions.

Most of these are empirical quantities, and none should depend on whether one supports or opposes a wealth tax.

It takes 5 to 10 minutes. Answers are anonymous unless you leave an email.

* Indicates required question

Do you consider yourself an expert in tax or fiscal policy? *

- ○ Yes
- ○ Somewhat
- ○ No

Before opening this survey, had you looked at news sources, web pages or academic studies discussing estimates of Proposition 40, the Billionaire Wealth Tax Proposal in California? *

- ○ Yes
- ○ No
- ○ Not sure

Your email (optional)

If you leave one, your answers are also reported in a separate analysis of the respondents who gave an email.

Your answer

The inputs

For each input, pick the value you find more credible.

1. The income tax lost when billionaires leave

The share of billionaires' income tax lost when people leave: which is more credible? *

- ○ A. 45% of all the income tax that billionaires pay: more than the 29% of their wealth reported to have left, because income tax need not leave in step with wealth and some moves go unreported.
- ○ B. 5% of all the income tax that billionaires pay: the same as the 5% of their wealth assumed to leave, because the largest of the billionaires reported to have left pay little income tax for their wealth.
- ○ About equally credible
- ○ Unsure

Why? (optional)

Your answer

How long the lost income tax is counted: which is more credible? *

- ○ A. For ever: billionaires who leave are assumed not to return.
- ○ B. 5 years, about the payment period, treating the departures as temporary.
- ○ About equally credible
- ○ Unsure

Why? (optional)

Your answer

Billionaires' yearly California income tax: which is more credible? *

- ○ A. $4.55 billion a year: from California tax data, taking the billionaires to be among the state's 212 to 500 highest earners.
- ○ B. $3.06 billion a year: from California tax data, taking the billionaires to have about half the income of the state's highest earners, as a study of top wealth holders finds.
- ○ About equally credible
- ○ Unsure

Why? (optional)

Your answer

2. Discounting the revenue for court risk

Discounting the revenue for the chance that courts strike the tax down: which is more credible? *

- ○ A. No discount: the tax is assessed at face value, as written.
- ○ B. A discount: the wealth tax collected is multiplied by the chance the tax survives in court, put at 50%.
- ○ About equally credible
- ○ Unsure

Why? (optional)

Your answer

3. Billionaires who moved out in time

Billionaires who moved out of California before the 1 January 2026 residency date: which is more credible? *

- ○ A. About 29% of the wealth: seven billionaires (Page, Thiel, Hankey, Kalanick, Brin, Spielberg and Sacks) are counted as having moved in time, on public reports of their moves.
- ○ B. 5% of the wealth: an allowance for billionaires who moved in time, with no one counted by name without evidence of a change of legal residence before the date.
- ○ About equally credible
- ○ Unsure

Why? (optional)

Your answer

The next 5 questions in total explain up to 3% of the difference between the two analyses. Answering them makes the survey on average 55% longer. Answering them is optional. *

- ○ Yes, show me the five (goes to the page “Five smaller inputs (optional)”)
- ○ No, skip them (goes to the page “Anything else (optional)”)

Five smaller inputs (optional)

Each of these moves the estimate only a little. Answer any you like, or none.

Wealth that escapes the tax through avoidance and evasion, beyond moving away: which is more credible?

- ○ A. 5%: a small allowance argued from enforcement.
- ○ B. None.
- ○ About equally credible
- ○ Unsure

The billionaires' wealth the tax applies to: which is more credible?

- ○ A. The Forbes list of 1 July 2026: 250 California billionaires worth $2,307 billion, or $2,157 billion without the non-US citizens asked about next.
- ○ B. The Forbes list of 2025: 212 California billionaires, grown 7% to the 31 December 2026 valuation date: $2,027 billion.
- ○ About equally credible
- ○ Unsure

Non-US-citizen residents: which is more credible?

- ○ A. Taxed: the $150 billion of wealth of California billionaires who are not US citizens is in the base.
- ○ B. Not taxed: the wealth of California billionaires who are not US citizens is left out of the base.
- ○ About equally credible
- ○ Unsure

Real-estate deduction: which is more credible?

- ○ A. No deduction: directly held real estate stays in the base.
- ○ B. A deduction: directly held real estate, about 0.5% of the base, is taken out of it.
- ○ About equally credible
- ○ Unsure

Extra income tax from assets sold to pay the tax: which is more credible?

- ○ A. Counted: one third of the tax is paid by selling assets, and the gains on them are taxed.
- ○ B. Not counted: no extra income tax from selling assets is included.
- ○ About equally credible
- ○ Unsure

Why? (optional)

Your answer

PAGE 4 OF 4

Anything else (optional)

Is there an input missing from this list that you think matters?

Your answer

Do you support Proposition 40?

- ○ Yes, I support it
- ○ No, I oppose it
- ○ Undecided
- ○ I prefer not to say

AFTER SUBMIT: THE CONFIRMATION MESSAGE

Thank you for taking the time to answer. All these answers will help us get closer to agreeing on common facts to inform important policy discussions.

See how the two estimates compare: tinyurl.com/prop40-explorer
About the project: tinyurl.com/prop40-opa

Appendix B.2. The treatment form (forms 4 and 6)

Printed from the form's own definition as Google Forms served it in preview on 18 October 2026. Each page of the form starts at a purple rule. Respondents see the text exactly as below, one page at a time.

PAGE 1 OF 5

What are the actual quantities? Proposition 40, input by input

Proposition 40 would tax Californians worth $1 billion or more once, at 5% of their net worth. Estimates of what it is worth to the state depend on a handful of inputs. Each question shows one input and two values that have been proposed for it. Tell us which value you find more credible. Your answers will help us get closer to agreeing on common facts to inform important policy discussions.

Most of these are empirical quantities, and none should depend on whether one supports or opposes a wealth tax.

It takes 5 to 10 minutes. Answers are anonymous unless you leave an email.

* Indicates required question

Do you consider yourself an expert in tax or fiscal policy? *

- ○ Yes
- ○ Somewhat
- ○ No

Before opening this survey, had you looked at news sources, web pages or academic studies discussing estimates of Proposition 40, the Billionaire Wealth Tax Proposal in California? *

- ○ Yes
- ○ No
- ○ Not sure

Your email (optional)

If you leave one, your answers are also reported in a separate analysis of the respondents who gave an email.

Your answer

How the two estimates differ

Two academic studies estimate what Proposition 40 is worth to California:
• Boll, Saez and Zucman (BSZ, August 2026): +$106.8 billion.
• Jaros and Rauh (JR, September 2026): –$39.0 billion.

The gap is $145.8 billion. Both are on one measure: the wealth tax collected, plus the extra income tax from assets sold to pay it, minus the income tax lost when people leave.

Two parts explain 83% of the gap: the income tax lost when billionaires leave, and the chance the tax survives in court. Below, how much of the gap each part explains, with each study's values for the first three.

1. The income tax lost when billionaires leave: the share lost, how long it is counted, and billionaires' yearly income tax, together
 $81.3 billion, 56% of the gap
BSZ: 5% of all the income tax that billionaires pay is lost; counted for 5 years; $3.06 billion a year.
JR: 45% of all the income tax that billionaires pay is lost; counted for ever; $4.55 billion a year.

2. Discounting the revenue for court risk
 $39.4 billion, 27% of the gap
BSZ: no discount.
JR: a 50% discount.

3. Billionaires who moved out in time
 $29.9 billion, 21% of the gap
BSZ: no billionaire counted by name (its allowance for 5% of the wealth moving away in time is in part 4).
JR: seven billionaires counted on public reports of their moves.

4. Other inputs together: the wealth on the list, avoidance, and three smaller inputs
 +$4.9 billion, offsetting 3% of the gap

The income tax lost is shown as one amount because its three inputs multiply each other. For the same reason, each amount depends on the order in which the inputs are changed.

The questions that follow ask which study's value of each input you find more credible.

The inputs

For each input, pick the value you find more credible.

1. The income tax lost when billionaires leave

The share of billionaires' income tax lost when people leave: which is more credible? *

- ○ JR: 45% of all the income tax that billionaires pay: more than the 29% of their wealth reported to have left, because income tax need not leave in step with wealth and some moves go unreported.
- ○ BSZ: 5% of all the income tax that billionaires pay: the same as the 5% of their wealth assumed to leave, because the largest of the billionaires reported to have left pay little income tax for their wealth.
- ○ About equally credible
- ○ Unsure

Why? (optional)

Your answer

How long the lost income tax is counted: which is more credible? *

- ○ JR: For ever: billionaires who leave are assumed not to return.
- ○ BSZ: 5 years, about the payment period, treating the departures as temporary.
- ○ About equally credible
- ○ Unsure

Why? (optional)

Your answer

Billionaires' yearly California income tax: which is more credible? *

- ○ JR: $4.55 billion a year: from California tax data, taking the billionaires to be among the state's 212 to 500 highest earners.
- ○ BSZ: $3.06 billion a year: from California tax data, taking the billionaires to have about half the income of the state's highest earners, as a study of top wealth holders finds.
- ○ About equally credible
- ○ Unsure

Why? (optional)

Your answer

2. Discounting the revenue for court risk

Discounting the revenue for the chance that courts strike the tax down: which is more credible? *

- ○ BSZ: No discount: the tax is assessed at face value, as written.
- ○ JR: A discount: the wealth tax collected is multiplied by the chance the tax survives in court, put at 50%.
- ○ About equally credible
- ○ Unsure

Why? (optional)

Your answer

3. Billionaires who moved out in time

Billionaires who moved out of California before the 1 January 2026 residency date: which is more credible? *

- ○ JR: About 29% of the wealth: seven billionaires (Page, Thiel, Hankey, Kalanick, Brin, Spielberg and Sacks) are counted as having moved in time, on public reports of their moves.
- ○ BSZ: 5% of the wealth: an allowance for billionaires who moved in time, with no one counted by name without evidence of a change of legal residence before the date.
- ○ About equally credible
- ○ Unsure

Why? (optional)

Your answer

The next 5 questions in total explain up to 3% of the difference between the two analyses. Answering them makes the survey on average 55% longer. Answering them is optional. *

- ○ Yes, show me the five (goes to the page “Five smaller inputs (optional)”)
- ○ No, skip them (goes to the page “Anything else (optional)”)

Five smaller inputs (optional)

Each of these moves the estimate only a little. Answer any you like, or none.

Wealth that escapes the tax through avoidance and evasion, beyond moving away: which is more credible?

- ○ BSZ: 5%: a small allowance argued from enforcement.
- ○ JR: None.
- ○ About equally credible
- ○ Unsure

The billionaires' wealth the tax applies to: which is more credible?

- ○ BSZ: The Forbes list of 1 July 2026: 250 California billionaires worth $2,307 billion, or $2,157 billion without the non-US citizens asked about next.
- ○ JR: The Forbes list of 2025: 212 California billionaires, grown 7% to the 31 December 2026 valuation date: $2,027 billion.
- ○ About equally credible
- ○ Unsure

Non-US-citizen residents: which is more credible?

- ○ BSZ: Taxed: the $150 billion of wealth of California billionaires who are not US citizens is in the base.
- ○ JR: Not taxed: the wealth of California billionaires who are not US citizens is left out of the base.
- ○ About equally credible
- ○ Unsure

Real-estate deduction: which is more credible?

- ○ BSZ: No deduction: directly held real estate stays in the base.
- ○ JR: A deduction: directly held real estate, about 0.5% of the base, is taken out of it.
- ○ About equally credible
- ○ Unsure

Extra income tax from assets sold to pay the tax: which is more credible?

- ○ BSZ: Counted: one third of the tax is paid by selling assets, and the gains on them are taxed.
- ○ JR: Not counted: no extra income tax from selling assets is included.
- ○ About equally credible
- ○ Unsure

Why? (optional)

Your answer

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Anything else (optional)

Is there an input missing from this list that you think matters?

Your answer

Do you support Proposition 40?

- ○ Yes, I support it
- ○ No, I oppose it
- ○ Undecided
- ○ I prefer not to say

AFTER SUBMIT: THE CONFIRMATION MESSAGE

Thank you for taking the time to answer. All these answers will help us get closer to agreeing on common facts to inform important policy discussions.

See how the two estimates compare: tinyurl.com/prop40-explorer
About the project: tinyurl.com/prop40-opa